

Abhinav Gupta

Curriculum Vitae

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EDUCATION

University of Southern California Los Angeles, CA
B.S. Computer Science, B.A. Cognitive Science Honors, AI Applications Specialization May 2026

- **Major GPA:** Computer Science 3.72, Cognitive Science 3.81
- **Relevant Coursework:** Natural Language Processing, Machine Learning, Linear Algebra, Probability and Statistics, Language and Mind, Human Language and Technology, Cognitive Processes

RESEARCH EXPERIENCE

Undergraduate Researcher, Interaction Lab December 2024 – Present
University of Southern California Advisor: Dr. Maja Matarić

- Investigating whether large language models can infer human actions and latent motives in strategic decision making, benchmarking against human behavioral data and statistical baselines
- Evaluating model prediction performance in the Social Prediction Game (SPG) and Inspection Game (IG), where LLMs observe simulated agents with rule-based motives such as greed and risk aversion
- Found that LLMs fail to adapt predictions over time unlike humans who develop mental models based on opponents' move patterns, revealing gaps in cross-game generalization that suggest fundamental differences in theory of mind reasoning
- Co-authored “Can Large Language Models Infer Human Actions and Motives in Strategic Decision Making,” which has been submitted to AAAI 2026.

Undergraduate Researcher, Language Development Lab December 2023 – Present
University of Southern California Advisor: Dr. Toben H. Mintz

- Designed and led a behavioral study with 250+ participants examining sensorimotor associations for pseudowords across eleven perceptual and motor modalities
- Constructed a controlled pseudoword stimulus set using Wuggy generation and computational model-based sensorimotor projections to isolate sublexical sound–meaning cues
- Performed statistical modeling and visualization, with preliminary results suggesting humans and models align on interoceptive and auditory modalities but diverge strongly on olfactory and haptic dimensions
- Co-authored work on sound–meaning mapping submitted to ACL 2026

Undergraduate Researcher, GLAMOR Lab August 2023 – Present
University of Southern California Advisor: Dr. Jesse Thomason

- Developed SENSE (Sensorimotor Embedding Norm Scoring Engine), a system mapping embeddings (Word2Vec, GloVe, BERT CLS) into the 11-dimensional Lancaster sensorimotor norms
- Benchmarked grounding across embeddings using baseline, KNN, and neural architectures, revealing that distributional embeddings encode gustatory information more reliably than other modalities
- Applied SENSE to downstream tasks including emoji grounding, revealing that emoji embeddings capture sensorimotor dimensions beyond visual similarity
- Co-authored work on sensorimotor grounding submitted to ACL 2026 and presented at the Southern California NLP Symposium 2024

Undergraduate Researcher, CAIS++ September 2022 – December 2023
USC & Universidad Nacional de Colombia Advisor: Dr. Robin Jia

- Partnered with the Muysccubun revitalization group to build NLP resources for Muisca (Chibcha), an Indigenous language with no prior digital corpus

- Created the first parallel Muisca–Spanish corpus of 6,000 sentence pairs from colonial manuscripts using custom scraping, parsing, and alignment algorithms tailored to handle historical orthographic variations
- Fine-tuned mT5 for low-resource machine translation, outperforming other endangered language MT systems despite using a smaller corpus, demonstrating the effectiveness of our preprocessing methods
- Built an interactive web-based educational tool to support community-led language revitalization and presented this work at the Southern California NLP Symposium 2023

Research Assistant, SAIL

University of Southern California

August 2021 – May 2022

Advisor: Dr. Shrikanth Narayanan

- Developed preprocessing pipelines and annotated multimodal video datasets combining facial, audio, and textual signals for emotion recognition experiments
- Reproduced baseline emotion recognition models and ran ablation experiments on bias mitigation techniques
- Identified lower accuracy for certain demographic subgroups, revealing bias patterns stemming from training data sourced from American television shows with limited demographic diversity

PUBLICATIONS & PRESENTATIONS

- [Words that make SENSE: Sensorimotor Norms in Learned Lexical Token Representations](#). **Abhinav Gupta**, Toben H. Mintz, Jesse Thomason *Findings of Association for Computational Linguistics*, 2026.
- [Can Large Language Models Infer Human Actions and Motives in Strategic Decision Making?](#) Kaleen Shrestha, Harish Dukkupati, **Abhinav Gupta**, Zhonghao Shi, Maja Matarić. *Findings of Association for Computational Linguistics*, 2026.
- [Extracting Sensorimotor Associations of Phrases](#). **Abhinav Gupta**, Jesse Thomason. *Southern California NLP Symposium*, 2024.
- Endangered Language Preservation using Machine Translation. Aryan Gulati, Leslie Moreno, **Abhinav Gupta**, Aditya Kumar. Second-best poster award, *USC ShowCAIS Research Showcase*, 2023.
- Creating a Parallel Corpus for a Low-Resource, Indigenous Language: Muisca-to-Spanish. Aryan Gulati, Leslie Moreno, Aditya Kumar, **Abhinav Gupta**. *Southern California NLP Symposium*, 2023.

HONORS, SCHOLARSHIPS & AWARDS

- USC Renaissance Scholar Prize March 2025
- USC Discovery Scholar Distinction February 2025
- USC Dornsife Continuing Student Scholarship August 2025 – Present
- Center for Undergraduate Research in Viterbi Engineering Fellowship May 2025 – August 2025
- USC Cognitive Science Honors August 2024 – Present
- USC Provost Fellowship May 2024 – August 2024
- Undergraduate Research Associates Program Scholar August 2023 – May 2024
- Second-best poster presentation, USC ShowCAIS April 2023
- Dean’s List (Dornsife College) Spring 2022, Fall 2022, Fall 2024, Spring 2025
- Dean’s List (Viterbi School of Engineering) Fall 2021, Spring 2022, Fall 2022, Fall 2024

TEACHING & MENTORING EXPERIENCE

Math Peer Mentor

USC Viterbi MESA Program

August 2025 – Present

Los Angeles, CA

- Mentor first-year students in calculus, supporting problem-solving skills and effective study habits
- Conduct program outreach through social media, classroom visits, and collaboration with student orgs

Course Producer & Peer Mentor

USC Viterbi School of Engineering

January 2022 – December 2024

Los Angeles, CA

- Supported CSCI 270, CSCI 170, and CSCI 102 through tutoring, office hours, and technical guidance

- Mentored more than 100 students and contributed to inclusive learning in large-enrollment courses
- Assisted with designing assignments, curating assessments, and refining instructional materials

Computer Science Tutor

September 2021 – January 2022

Viterbi Impact Program

Los Angeles, CA

- Taught foundational programming concepts to middle school students and teachers in community STEM programs
- Led curriculum discussions and coordinated with program instructors to improve lesson flow

INDUSTRY EXPERIENCE

Software Development Engineering Intern

May 2023 – August 2023

Amazon Inc.

Culver City, CA

- Designed an internal automation tool using Java, Python, and AWS
- Reduced recurring manual workload by 10 hours per week through end-to-end automation

Full Stack Developer & AI Engineer

May 2022 – December 2022

Bacpoint, Inc.

Los Angeles, CA

- Built the full mobile application stack for *Ethos*, a mindful consumption app
- Led a team of five engineers developing a predictive AI model to estimate alcohol intake from facial signals

TECHNICAL SKILLS

- **Research Methods:** Experimental design, human subject studies, statistical modeling, psycholinguistic analysis, behavioral data analysis, data visualization
- **Machine Learning & NLP:** PyTorch, TensorFlow, Hugging Face, spaCy, scikit-learn, representation learning, LLM evaluation, sequence modeling
- **Programming:** Python, R, Java, C++, JavaScript, SQL
- **Data & Tools:** NumPy, pandas, Jupyter, Git, Docker, Linux, LaTeX
- **Web Development:** React, Node.js, Django, AWS

SERVICE & LEADERSHIP

Software Developer & Volunteer

September 2022 – July 2024

Code the Change

Los Angeles, CA

- Developed software for non-profit partners focused on accessibility, safety, and community support
- Contributed as a full-stack developer to a community platform for assault survivors, applying computing to sensitive social challenges

Curriculum Lead & Project Member

September 2021 – January 2024

USC Center for AI in Society (CAIS++)

Los Angeles, CA

- Led a team of seven students designing and teaching a semester-long curriculum on AI and machine learning for social good
- Provided technical mentorship to NLP project teams, including work that was later presented to the broader USC and SoCal AI communities
- Participated in interdisciplinary discussions on aligning AI systems with social impact